

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Mechanical Semester :II Subject Code & Name:BTMEC602, Machine Design - II Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Derive stribeck equation for static load carrying capacity of bearing	CO3	06
B)	The radial Fr=2500N for first five revolution reduces to 1500N for next ten revolution. Variation of load is repetitive; The expected life of the bearing is 25 million revolutions. Determine dynamic load carrying capacity of bearing	CO4	06
C)	Discuss in detail about Viscosity and its measurement of lubricant	CO2	06
Q.2	Solve Any Two of the following.		
A)	Compare Hydrostatic and Hydrodynamic bearing	CO3	06
B)	What is Destructive pitting and Scoring failure of gear	CO2	06
C)	Discuss virtual number of teeth for helical gear with neat sketch	CO4	06
Q. 3	Solve Any One of the following.		
A)	A spur gear pair is to transmit 8KW power at 750 rpm to another machine at 300 rpm. Material for both the gears are same having ultimate tensile strength is 620 MPa. Surface hardness is 330 BHN. Pinion has 20 teeth, module is 5mm and face width is 60 mm service factor is 1.5 mm and load concentration factor is 1.4. Determine the factor of safety in bending and wear a) by considering the velocity factor accounts for dynamic load b) by considering dynamic load using Buckingham equation. Velocity factor = $3/(3+V)$, Tooth profile error is 0.05 mm, tooth form factor = 0.111. $E=2.1 \times 10^5$ MPa.	CO5	12
B)	Write short note on a) Thermal consideration of worm and worm wheel b) Selection of material for worm and worm wheel	CO5	12

C)	A bevel gear pair consists of 20 teeth on pinion meshing with 44 teeth gear, pinion and gear having ultimate tensile strength are 680 MPa and 600 MPa respectively. Pinion transmits 15 KW at 1500 rpm. Service factor 1.75, factor of safety 1.5, surface hardness 350 BHN for both gears. Assume the velocity factor accounts for dynamic load. $C_v = 5.6/(5.6 + \sqrt{v})$. Design a gear pair	CO5	12
Q.4	Solve Any Two of the following.		
A)	The leather belt drives transmit 15 KW of power by open belt. The center distance between the pulleys is twice the diameter of bigger pulley. The belt should operate at velocity of 20 m/s approx. and the stress in the belt should be 2.25 MPa. The thickness of belt is 5 mm. Calculate the diameter of pulleys the length and width of the belt and the belt tension.	CO5	06
B)	State polygonal effect in chain drive with neat sketch	CO4	06
C)	What is stresses in disc and armed flywheel	CO3	06
Q. 5	Solve Any One of the following.		
A)	A centrifugal clutch transmits 25 KW power at 900 rpm. The speed at which engagement begins is 70% of the running speed. The inner radius of drum is 15 cm and the radial distance of centre of gravity of shoe beam the axis of rotation of the spider in engaged position is 12 cm. Determine mass of each shoe. Take number of shoes as four and coefficient of friction is 0.25 and intensity of pressure is 0.1 MPa.	CO4	12
B)	Discuss Shoe brake for line of action passes through fulcrum of lever and below and above at a distance a w.r.t fulcrum	CO4	12
C)	A cylinder diameter is 200 mm closed on both its ends is withstanding internal pressure of 20 MPa. Ultimate and yield strength is 440MPa and 240MPa respectively and factor of safety is 1.5 and poissons ration is 0.3. Determine thickness of cylinder based on maximum principal stress theory, maximum principal strain theory, maximum shear stress theory and maximum distortion energy theory.	CO6	12
*** End ***			

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